Heart Failure

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Definition

Heart failure is a clinical and pathophysiological syndrome that results from inability of the heart to function adequately to meet the metabolic demands of the body.

Classification of Functional Status of Heart Failure

The modified Ross classification is most widely accepted and used amongst the many available classifications for pediatric heart failure **(Table1)**.

TABLE 1: Modified Ross classification of heart failure.

Class I	No symptoms/limitations
Class II	Mild tachypnea/sweating during feeds in infants/dyspnoea on exertion in older children but no growth failure
Class III	Significant tachypnea or sweating during feeds/marked dyspnoea on exertion/prolonged feeding time with growth failure
Class IV	Symptoms (tachypnoea, retractions, grunting and sweating) even at rest with growth failure

Etiology

The most important step in managing heart failure in a child is to determine the exact cause **(Table 2)**.

TABLE 2: Classification of causes of heart failure as determined by underlying mechanism.

Category	*Selected Specific Conditions*
Shunt lesions	☑ Large post-tricuspid shunts (VSD, PDA, AP window) ☑ Selected pre-tricuspid shunts (small proportion of ASD patients) ☑ Combinations (Shunt lesion + Obstruction; ASD with mitral valve disease) ☑ Mixed Shunts (AV septal defects) ☑ Cyanotic heart disease with increased pulmonary blood flow ☑ Large systemic AV malformations (Vein of Galen malformation)
Right heart pathology	☑ Tricuspid valve disease—severe regurgitation or stenosis ☑ Severe outflow obstruction—severe stenosis involving infundibulum, pulmonary valve, main pulmonary artery or branch pulmonary arteries ☑ Long standing severe pulmonary regurgitation with RV dilation and dysfunction ☑ Pulmonary hypertension ☑ Isolate right ventricular dysfunction ☑ Restrictive physiology
Left heart pathology	☑ Mitral valve disease ☑ Aortic valve disease ☑ Outflow obstruction ☑ Systemic hypertension ☑ Left ventricular systolic or diastolic dysfunction
Rhythm disorders	☑ Tachyarrhythmias (e.g., longstanding supraventricular tachycardia) ☑ Bradyarrhythmias (e.g., congenital heart block)
Ischemic	☑ Premature coronary artery disease ☑ Congenital coronary anomaly (ALCAPA) ☑ Postoperative coronary occlusion ☑ Post-Kawasaki disease
Acquired conditions affecting myocardium	☑ Myocarditis ☑ Drug- or toxin-related (e.g., Cancer chemotherapy, iron overload) ☑ Anemia ☑ Thyrotoxicosis and thyroid deficiency
Intrinsic myopathies	☑ Metabolic ☑ Mitochondrial ☑ Neuromuscular ☑ Familial ☑ Idiopathic
Restrictive heart disease	☑ Constrictive pericarditis ☑ Endomyocardial fibrosis ☑ Idiopathic restrictive cardiomyopathy

(ALCAPA: anomalous coronary artery from pulmonary artery; AP: Aortopulmonary; ASD: atrial septal defect; PDA: patent ductus arteriosus; VSD: ventricular septal defect)

It is essential to identify co-morbidities and differentiate cardiac from noncardiac causes of heart failure. The tests help that guide the diagnosis and management are listed in **Table 3**.

TABLE 3: Suggested diagnostic work for heart failure (HF).

Tests	***Utility***
Complete blood count	Identifying sepsis, anemia
Arterial blood gas with lactate	Lactic acidosis—as a marker of tissue perfusion and helps monitor response to treatment; It is also elevated in specific inborn errors of metabolism
Electrolytes and urea, creatinine	Elevated urea, creatinine may indicate decompensated HF or may result from medication side effects. Electrolyte imbalance is a common association of HF and diuretic use. Hypocalcemia can cause ventricular dysfunction leading to HF
Liver function test	Elevated bilirubin, liver enzymes and prolonged prothrombin time points towards congestive hepatopathy. Hypoalbuminemia points to chronic HF and poor nutrition
Thyroid function test	Thyroid hormone imbalance could be a primary cause or may lead to worsening of symptoms
Brain natriuretic peptide (BNP)	It helps differentiate HF from respiratory disease. Useful in monitoring response to therapy
Cardiac enzymes (troponin I, T, CKMB)	In suspected cases of myocarditis
Viral panel	Suspected viral myocarditis
Chest X-ray	Information on cardiac silhouette, pulmonary vasculature, pulmonary artery dilatation and associated skeletal abnormalities
ECG	Diagnosis of treatable causes of heart failure such as persistent tachyarrhythmia, ALCAPA and, hypocalcemia. Other specific causes such as Pompe's disease, specific forms of cardiac muscle involvement in muscular dystrophy have ECG manifestations.
Echocardiogram	Critically important to accurate diagnosis and tailoring response to therapy
Cardiac CT/MRI	Required whenever echocardiography windows are suboptimal. MRI allows for a comprehensive physiological assessment in addition to anatomic details. CT allows high-resolution images of cardiac and extra-cardiac structures
Cardiac catheterization	Allows therapeutic interventions to manage heart failure. Examples include closure of shunts and fistulae, balloon dilation of critically narrow valves, balloon dilation or stenting of vessels. Diagnostic cardiac catheterization is particularly useful for quantifying pulmonary artery pressure and identifying the cause of pulmonary hypertension
Six-minute Walk test	Helpful in ascertaining functional status and response to therapy

(ALCAPA: anomalous coronary artery from pulmonary arter)

The main goals of HF therapy are to correct the underlying cause, reduce associated morbidity and mortality, improve functional status and improve quality of life.

Fluid Management

- ☑ **Fluid management:** Fluid intake is restricted in acute HF where there is lung congestion, peripheral edema despite diuretics or when there is hyponatremia.
- ☑ **Rest and restriction of activity:** Nasogastric feeding is helpful for infants in acute severe HF. Mild-to-moderate aerobic activity should be encouraged in older children with chronic compensated HF.
- ☑ **Correction of anemia:** Blood transfusion is generally reserved for severe anemia (Hb 7 g/dL or less).
- ☑ **Nutrition:** In infants calorie intake of 120–150 kcal/kg/day is recommended with a fluid intake of 100 mL/kg/day. This is achieved by thickening of feeds or by adding fat (coconut oil/medium chain triglyceride). In older children increase protein content of diet while optimizing the fat and carbohydrate intake. Dietary supplementation of calcium and vitamin D_3 should be encouraged. Dietary restriction of sodium is generally not recommended in children unless there is severe edema unresponsive to diuretic therapy. Supplementary oxygen may be necessary when there is respiratory distress.

Drug Therapy Goals

- ☑ Reduce congestion in lungs and other organs
- ☑ Increase cardiac output with improved end-organ perfusion:
  - Decrease systemic vascular resistance without decreasing tissue perfusion whenever physiologically appropriate.
  - Reduction in systemic vascular resistance is not safe when there is dynamic or fixed left ventricular outflow obstruction.
- ☑ Optimize the preload without causing intra-arterial volume depletion.
- ☑ Potentially delay disease progression through improved ventricular remodeling.

Diuretics

- ☑ It is the mainstay of therapy and reduces systemic and pulmonary venous congestion.
- ☑ Furosemide—usual dose is 1–2 mg/kg q 6–12 hours. In acute heart failure, it can be used as infusion at a dose of 1–3 mg/kg/day. In case of refractoriness addition of thiazide diuretics such as metolazone may be done. Spironolactone (Aldosterone antagonist) 0.5–1.5 mg/kg q 12 hours decreases potassium loss and may prevent myocardial fibrosis.
- ☑ It is essential to monitor for hypovolemia, hypotension, hypokalemia, hyponatremia metabolic alkalosis and renal failure.

Management

Vasodilators

- ☑ These are administered in HF only when the altered physiology is likely to benefit from a reduction in vascular resistance such as in systolic ventricular dysfunction with no outflow track obstruction and acute left-sided valve regurgitation.
- ☑ *Angiotensin converting enzyme inhibitor:* Enalapril—most widely used—0.1–0.5 mg/kg/day in 2–3 divided doses. Side effects include cough, hypotension and renal failure. The first dose of 0.1 mg serves as a 'test dose' and helps avoid serious hypotension.
- ☑ Angiotensin receptor blocker: Losartan 0.75–1.4 mg/kg/day for those patients who do not tolerate Enalapril.
- ☑ *Nitroglycerine/nitroprusside:* Acute setting for afterload reduction. This has largely been replaced by milrinone and levosimendan that are essentially vasodilators with an additional inotropic effect (inodilators).

Inotropes

Digoxin is an orally administered cardiac glycoside—10 µg/kg/day orally, serum levels to be maintained between 0.5 and 0.9 nmol/L. Intravenous Dobutamine, Dopamine and Epinephrine are common choices of inotropes, while Milrinone and Levosimendan are used as inodilators.

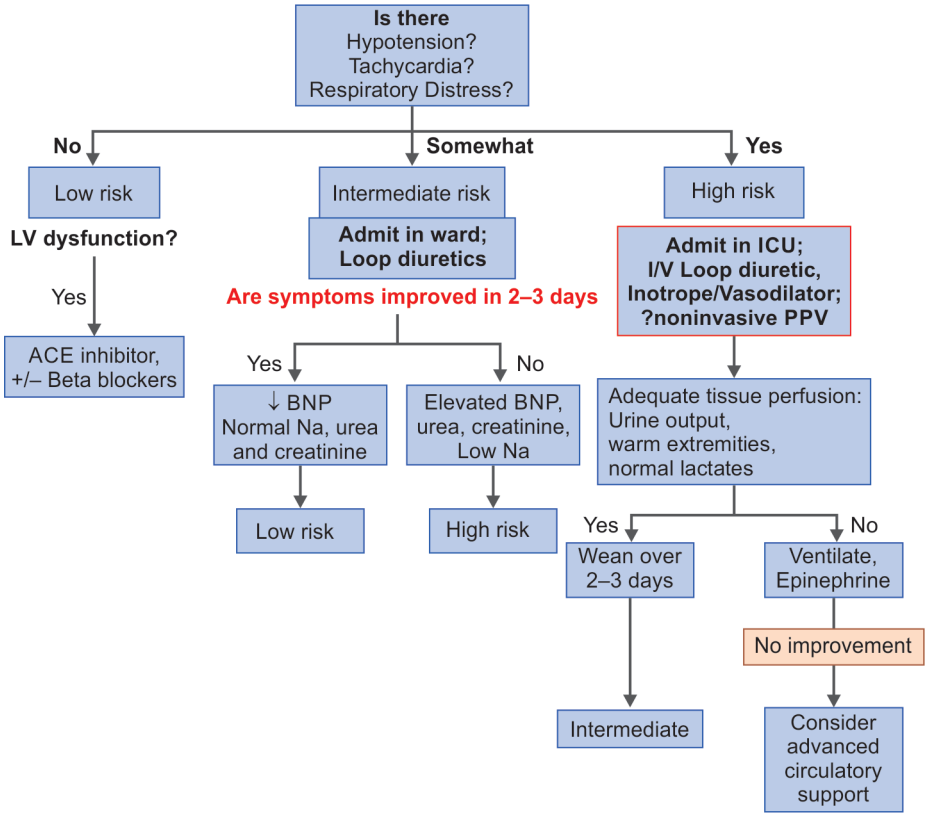
Beta Blockers

Reduces tachycardia and improves diastolic filling of heart. It is useful in mitral stenosis, hypertrophic obstructive cardiomyopathy and tachyarrhythmia related heart failure. Commonly used agents are non-selective betablockers such as propranolol or selective agents such as metoprolol, atenolol or bisoprolol. Carvedilol (0.1–0.8 mg/kg/day in three divided doses) has additional vasodilatory properties and is most commonly used for selected case of cardiomyopathy and for persistent ventricular dysfunction after cardiac surgery (such as after ALCAPA repair).

Newer Drugs

- ☑ Ivabradine—used to control heart rate with no negative inotropic effect. Combination of Neprilysin inhibitor (Sacubitril) and Valsartan is now increasingly used in children and results of clinical trials are awaited.
- ☑ A suggested management algorithm for acute heart failure is shown in **Flowchart 1**.

Flowchart 1: Management of heart failure.



Treatment of Advanced Pediatric Heart Failure and End-stage Heart Disease

Advanced therapies for chronic heart failure include cardiac resynchronization therapy (CRT), ventricular assist devices (typically as a bridge to cardiac transplantation) and heart transplant. These therapies are expensive and there are significant logistic challenges in the Indian setting.

Further Reading

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- ☑ Venkatesh S, Kumar RK, Heart Failure in Children. IAP specialty Series on Pediatric Cardiology, 3rd edition. Jaypee Brothers Medical Publishers, New Delhi; 2022. . pp. 351-76.

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